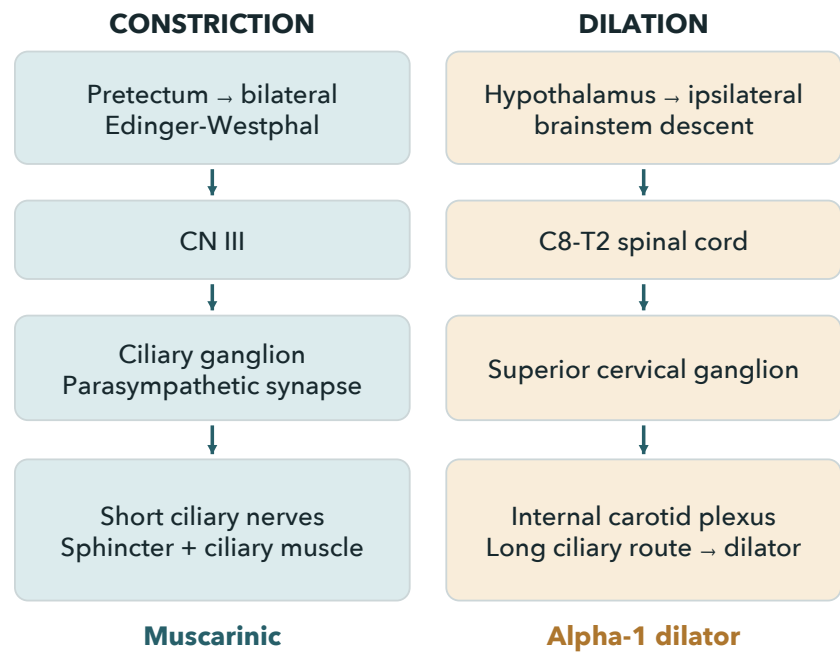


# Autonomic pupil circuits

Trace constriction and dilation separately, then compare both eyes.

## Two motor routes, different synapses

### A | Parasympathetic and sympathetic routes



Sympathetic descent is ipsilateral. Sensory and sympathetic fibres traverse the ciliary ganglion without synapsing.

### Targets and receptors

| Route            | Receptor           | Effect                                   |
|------------------|--------------------|------------------------------------------|
| Parasympathetic  | Muscarinic         | Sphincter and ciliary muscle contract    |
| Sympathetic      | Alpha-1 at dilator | Radial dilator contracts                 |
| Ciliary ganglion | Nicotinic          | Parasympathetic ganglion cells activated |

### Fast check

After the superior cervical ganglion, ocular fibres follow the internal carotid route; many facial sweating fibres follow the external carotid route. Sweating loss therefore varies with lesion level.

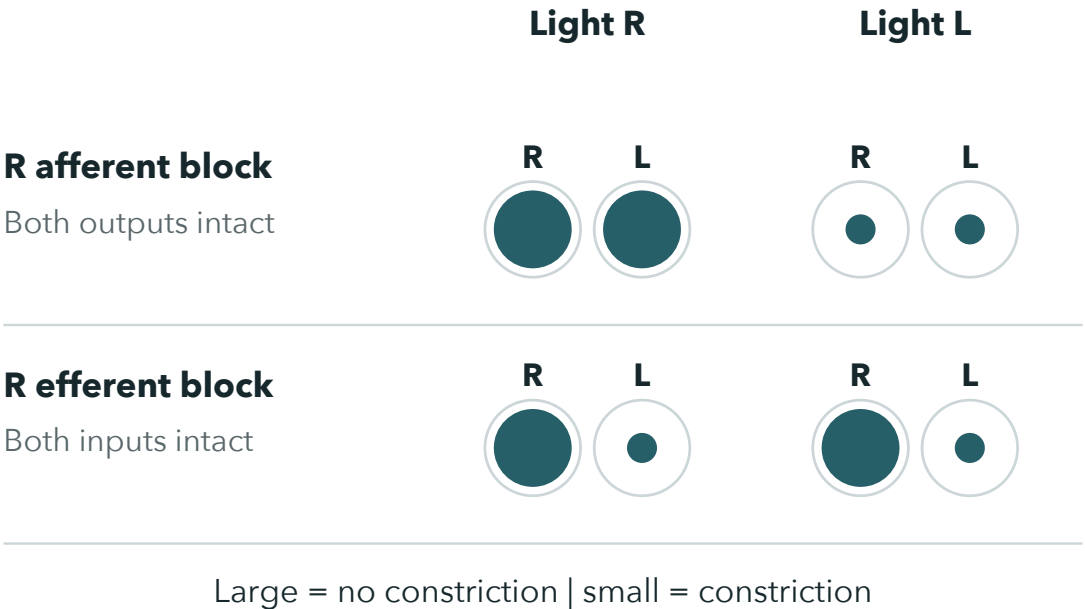
### Figure reference

A: pupil reflex and Horner pathways. Original schematic.

# Stimulus versus effector failure

## B | Read both pupils under both stimuli

### Responses to each light stimulus



Normal left constriction with right-eye illumination demonstrates usable right afferent input in the efferent example.

## Add the near response

| Pattern                               | Mechanism clue                                                        |
|---------------------------------------|-----------------------------------------------------------------------|
| Poor light, better near               | Different premotor drives can reach shared final effectors            |
| Slow near constriction and redilation | Postganglionic denervation/reinnervation can produce a tonic response |
| Stronger response to dilute agonist   | Denervation hypersensitivity supports the mechanism in context        |

## Fast check

Compare anisocoria in light and dark: a larger difference in light points towards a constrictor problem; a larger difference in dark points towards a dilator problem. This is a clue, not a complete diagnosis.

## Figure reference

B: pupil reflex, anisocoria and Adie references. Original response matrix.